The Bargmann–Segal Lattice: π-Free Discretization of the Harmonic Oscillator on S^5

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(Dated: April 10, 2026)

We construct a discrete graph encoding of the three-dimensional isotropic harmonic oscillator on the holomorphic sector of S^5 and verify by exact rational arithmetic that the resulting graph and its projection to physical energies contain no transcendental constants. The Bargmann–Segal transform [2, 4] maps the harmonic oscillator Hamiltonian to $\hbar\omega(\hat{N} + 3/2)$ on the Fock–Bargmann space, where $\hat{N}$ is the Euler number operator on holomorphic monomials. Restricting to the unit sphere $S^5 \subset \mathbb{C}^3$ identifies the eigenspaces with the symmetric SU(3) irreps $(N, 0)$, of dimension $(N + 1)(N + 2)/2$, which match the harmonic oscillator shell degeneracies exactly. The graph constructed from these states with dipole transition weights computed by exact `fractions.Fraction` arithmetic is bit-exactly π-free at $N_{\max} = 5$ (56 nodes, 165 edges, zero irrationals encountered). The single π in the entire ecosystem is the Gaussian normalization π^{-3} of the continuous Bargmann measure, which the discrete graph never invokes. The Coulomb problem on S^3 and the harmonic oscillator on S^5 are both natural sphere quotients, but they live on different spheres with different geometric character (Riemannian vs complex), have natural operators of different order (second vs first), yield spectra of different growth (quadratic vs linear), and demand projections of different type (nonlinear vs linear-affine). This structural asymmetry refines the exchange constant taxonomy [17]: calibration π is Coulomb-specific. It is the transcendental cost of mapping a quadratic Riemannian eigenvalue to an inverse-square physical energy, and it has no analog for the harmonic oscillator's linear complex-analytic projection.

I. INTRODUCTION

Fock's 1935 stereographic projection [1] maps the hydrogen Schrödinger equation onto the free Laplacian on the three-sphere S^3. Paper 0 [13] discretized this construction: nodes labeled by quantum numbers (n, l, m), edges from angular-momentum selection rules, and a graph Laplacian whose eigenvalues converge to those of the continuous Laplace–Beltrami operator. The kinetic constant $\kappa = -1/16$ then maps these continuum S^3 eigenvalues $n^2 - 1$ to physical Rydberg energies $E_n = -Z^2/(2n^2)$. Paper 7 [15] validated this S^3 reduction with eighteen independent symbolic proofs. Paper 18 [17] identified a transcendental cost in the projection: π enters through the spectral zeta values of S^3 as a *calibration* exchange constant.

The harmonic oscillator $V(r) = \frac{1}{2}m\omega^2r^2$ has a different dynamical symmetry group (SU(3) instead of SO(4)) and a qualitatively different spectrum (linear in shell index instead of inverse-square). The natural question is whether an analogous discretization exists for the harmonic oscillator, and if so, what role π plays.

We answer both questions in this paper. The discretization exists, via the Bargmann–Segal transform [2, 4] applied to the Hardy space $H^2(S^5)$. The graph is bit-exactly π-free at every finite $N_{\max}$. The single π in the entire construction appears in the Gaussian normalization π^{-3} of the continuous Bargmann measure, which is never used by the discrete graph. We further show that the Coulomb / harmonic oscillator asymmetry has a structural origin in the difference between Riemannian and complex natural geometries, and that this asymmetry refines the exchange constant taxonomy of Paper 18: *calibration* exchange constants are tied to second-order Riemannian operators with nonlinear projections, while linear projections from first-order complex-analytic operators introduce no calibration constants of any kind. The result has direct implications for the framing of the fine-structure constant Observation [14] and for the precision of nuclear shell-model qubit Hamiltonians built on harmonic oscillator single-particle bases [19].

The plan of the paper is as follows. Section II summarizes the Bargmann–Segal construction and its identification with the SU(3) symmetric representation theory. Section III constructs the graph and states the π-free certificate (Theorem 1). Section IV presents the Coulomb / HO structural asymmetry as the conceptual centerpiece (Theorem 2). Section V states the harmonic oscillator rigidity theorem (Theorem 3), the dual of the Fock projection rigidity theorem of Paper 23 [19]. Section VI discusses implications for the α Observation, for nuclear quantum simulation, and for the framework's universal / Coulomb-specific partition.

II. THE BARGMANN–SEGAL CONSTRUCTION

A. The Bargmann transform

The Bargmann transform [2] is the integral operator

$$[Bf](z) = \pi^{-3/4} \int_{\mathbb{R}^3} \exp\Big(-\tfrac{1}{2}(z^2 + x^2) + \sqrt{2}\, z \cdot x\Big)\, f(x)\, d^3x, \tag{1}$$

mapping $L^2(\mathbb{R}^3)$ unitarily onto the Fock–Bargmann space $F^2(\mathbb{C}^3)$ of holomorphic functions on $\mathbb{C}^3$ that are square-integrable against the Gaussian measure

$$d\mu(z) = \pi^{-3}\, e^{-|z|^2}\, d^6 z. \tag{2}$$

The harmonic-oscillator eigenstates $|N, l, m\rangle$ map under B to solid harmonic polynomials of total degree N in the complex coordinates z_1, z_2, z_3. The transform is unitary, so the eigenvalue equations carry across exactly.

B. The Euler operator

The number (Euler) operator on $F^2(\mathbb{C}^3)$ is

$$\hat{N} = \sum_{i=1}^{3} z_i \frac{\partial}{\partial z_i}, \tag{3}$$

which acts on the monomial $z_1^a z_2^b z_3^c$ with eigenvalue $a + b + c = N$. By multinomial counting, the eigenspace $\hat{N} = N$ has dimension

$$\dim\{z_1^a z_2^b z_3^c \,:\, a{+}b{+}c = N\} \;=\; \binom{N+2}{2} \;=\; \frac{(N+1)(N+2)}{2} \tag{4}$$

which matches the 3D harmonic oscillator shell degeneracy and the Weyl dimension formula for the symmetric irreducible representation $(N, 0)$ of SU(3).

Under the Bargmann transform, the harmonic oscillator Hamiltonian maps to

$$H_{\mathrm{HO}} \;=\; -\tfrac{1}{2m}\nabla^2 + \tfrac{1}{2}m\omega^2 r^2 \;\xleftrightarrow{B}\; \hbar\omega\big(\hat{N} + \tfrac{3}{2}\big), \tag{5}$$

so the spectrum is

$$E_N \;=\; \hbar\omega\big(N + \tfrac{3}{2}\big) \tag{6}$$

on the eigenspace of dimension $(N + 1)(N + 2)/2$. The HO Hamiltonian is a linear affine function of the Euler operator; the projection from the discrete eigenvalue to the physical energy involves only the universal scale $\hbar\omega$ and the additive zero-point shift 3/2.

The unitary equivalence of the Schrödinger and Bargmann pictures is rigorous and originates with Bargmann [2] and Segal [3]; the modern functional-analytic treatment is due to Hall [4], who extended the construction to arbitrary compact Lie groups.

C. The S^5 Hardy space and SU(3) symmetric representations

The unit sphere $S^5 = \{z \in \mathbb{C}^3 : |z| = 1\}$ carries a Cauchy–Riemann (CR) structure inherited from $\mathbb{C}^3$. The Hardy space $H^2(S^5)$ consists of boundary values of holomorphic functions on the unit ball $B^3 \subset \mathbb{C}^3$ that are square-integrable on S^5 with respect to the surface measure. Restriction to S^5 identifies $H^2(S^5)$ with the closure of the polynomial algebra $\mathbb{C}[z_1, z_2, z_3]$, with the natural SU(3) action by linear transformations of the coordinates.

The decomposition by polynomial degree gives an orthogonal direct sum

$$H^2(S^5) \;=\; \bigoplus_{N=0}^{\infty} \mathcal{H}_N, \qquad \dim \mathcal{H}_N = \frac{(N+1)(N+2)}{2}, \tag{7}$$

where $\mathcal{H}_N$ is the space of homogeneous degree-N holomorphic polynomials. Each $\mathcal{H}_N$ transforms as the SU(3) irreducible representation $(N, 0)$ — the symmetric tensor representation of rank N on the fundamental $\mathbf{3}$.

By the Borel–Weil theorem [7], this is the same data as the space of holomorphic sections of the line bundle $\mathcal{O}(N)$ over $\mathbb{CP}^2$, where $\mathbb{CP}^2 = \mathrm{SU}(3)/\mathrm{U}(2)$ is the natural quotient. The fibration

$$S^1 \;\longrightarrow\; S^5 \;\longrightarrow\; \mathbb{CP}^2 \tag{8}$$

is the structural analog of the Hopf fibration $S^1 \to S^3 \to S^2$ that played a central role in Paper 7 [15] and the α Observation of Paper 2 [14].

The CR Laplacian (Kohn Laplacian) $\Box_b$ on the holomorphic sector of S^5, the Dolbeault Laplacian on $\mathcal{O}(N) \to \mathbb{CP}^2$, and the Euler operator on the Bargmann monomials all act on the same SU(3) irreducibles $(N, 0)$ and produce spectra that are linear in the shell index N up to additive and multiplicative normalizations [5]. A genuinely spherical harmonic-oscillator analogue — one whose configuration space is the sphere itself and which, unlike the flat-space oscillator, does not factor into one-dimensional modes — has been studied independently in recent mathematical work [27], which identifies the two-dimensional case as critical. That construction is distinct from the flat-space isotropic oscillator discretized here via the S^5 Hardy sector; we note it only as an independent recent appearance of harmonic-oscillator dynamics with sphere symmetry. For the discretization in Section III we use the Bargmann description, which is the most elementary: the Euler operator is diagonal on the monomial basis, requires no differential geometry, and is amenable to immediate exact arithmetic.

III. THE BARGMANN GRAPH

A. Graph construction

a. Nodes. The nodes of the Bargmann graph at cutoff $N_{\max}$ are triples (N, l, m_l) with $N = 0, 1, 2, \ldots, N_{\max}$; for each N, the orbital angular momentum runs over $l = N, N{-}2, \ldots, 1$ or 0 (same parity as N); and for each l, the magnetic quantum number runs over $m_l = -l, \ldots, +l$. This is the SU(3) $\to$ SO(3) branching of the symmetric irrep $(N, 0)$. The total node count is

$$|V(N_{\max})| \;=\; \sum_{N=0}^{N_{\max}} \frac{(N+1)(N+2)}{2}. \tag{9}$$

At $N_{\max} = 5$ this gives $1+3+6+10+15+21 = 56$ nodes.

b. Edges. Edges connect states under SU(3) dipole selection rules:

$$\Delta N = \pm 1, \qquad \Delta l = \pm 1, \qquad \Delta m_l = 0, \pm 1. \tag{10}$$

These are the matrix elements of the position operator z_q ($q = 0, \pm 1$ in spherical components) between adjacent shells, which raise N by one and change l by ± 1 by the standard angular-momentum dipole rule.

c. Edge weights. For each allowed transition the edge weight is the squared matrix element

$$w_{(N,l,m_l)\to(N+1,l',m_l')} \;=\; R^2_{n_r,\,l\to l'} \;\times\; \left|C^{l'm_l'}_{l\,m_l;\,1\,q}\right|^2, \tag{11}$$

where the radial reduced matrix elements (in HO units $\hbar = m = \omega = 1$) are

$$\left|\langle n_r, l+1 \,|\, r \,|\, n_r, l\rangle\right|^2 = n_r + l + \tfrac{3}{2} = \tfrac{2n_r+2l+3}{2}, \tag{12}$$

$$\left|\langle n_r+1, l-1 \,|\, r \,|\, n_r, l\rangle\right|^2 = n_r + 1, \tag{13}$$

and the squared rank-1 Wigner $3j$ coefficients factorize into elementary rationals via the closed-form expressions of Edmonds [11] and Varshalovich [12]. Both factors in Eq. (11) are exact rational numbers, so the edge weight is exact rational.

d. Implementation. The reference implementation in `geovac/nuclear/bargmann_graph.py` uses Python's `fractions.Fraction` type throughout the construction. No floating-point arithmetic is used in the build path; the squared $3j$ tables are evaluated symbolically from the closed-form expressions, and the radial reduced matrix elements are constructed directly from the integer formulas above. Conversion to floating point occurs only when the dense Hamiltonian or adjacency is required for numerical eigensolvers.

B. The π-free certificate

Theorem 1 (Bargmann graph rationality)**.** *Let $G(N_{\max})$ be the Bargmann graph constructed in Section III with diagonal entries*

$$D_v \;=\; \mathrm{Fraction}(2N_v + 3,\, 2)\,, \tag{14}$$

in units of $\hbar\omega$, and adjacency weights given by Eq. (11) *computed in exact rational arithmetic. Then every diagonal entry and every nonzero adjacency weight is a rational number, and no transcendental constant appears in the construction at any finite $N_{\max}$.*

Proof. The diagonal entries are Fraction$(2N+3, 2)$ by construction (Section II B, Eq. (6)). The adjacency weights are products of two rational factors: the squared radial reduced matrix elements (which are rational linear functions of the integer quantum numbers n_r and l) and the squared rank-1 Wigner $3j$ coefficients (which are rational quotients of polynomials in the integer quantum numbers l, l', m_l, m_l' by the Edmonds [11] formulas). The product of two rationals is rational, and the construction terminates after finitely many additions and multiplications of rationals at any finite $N_{\max}$. □ □

a. Numerical verification. An independent run of the function `verify_pi_free(N_max=5)` in the reference implementation returns the certificate:

```
N_max:  5
n_nodes:  56
n_edges:  165
all_diagonal_rational:  True
all_adjacency_rational:  True
irrationals_encountered:  []
pi_free:  True
```

The 17 supporting unit tests are listed in `tests/test_bargmann_graph.py` and cover node enumeration, shell degeneracies, parity of l within each shell, the diagonal HO spectrum, the dipole selection rules, Hermiticity of the adjacency, exact rationality of every entry, and graph Laplacian positivity. All 17 pass.

b. Where the only π lives. The single π in the entire ecosystem appears in the Gaussian normalization π^{-3} of the continuous Bargmann measure Eq. (2). In Paper 18's exchange constant taxonomy [17], this is an *embedding* exchange constant: it quantifies the cost of representing the Bargmann Hilbert space as a measure space in 6 real dimensions, and it is fixed by the requirement that the Gaussian have unit total integral. The discrete graph never invokes this normalization. No analogous π enters the spectrum, the degeneracies, the selection rules, the Wigner $3j$ coefficients, or the radial matrix elements.

C. Hamiltonian and graph Laplacian roles

The HO Hamiltonian on the Bargmann graph is a diagonal matrix:

$$H_{\mathrm{HO}} \;=\; \hbar\omega\,\mathrm{diag}\big(N_v + \tfrac{3}{2}\big)_{v\in V}\,. \tag{15}$$

The eigenvalues are $\hbar\omega(3/2, 5/2, 7/2, 9/2, \ldots)$ with multiplicities $1, 3, 6, 10, \ldots = (N{+}1)(N{+}2)/2$. No off-diagonal terms are needed to recover the spectrum: it is simply the diagonal.

The adjacency matrix A encodes dipole transition amplitudes between HO states (electromagnetic absorption and emission lines). It plays no role in the construction of the spectrum; its role is to describe the connectivity structure that determines selection rules for transition processes.

This is qualitatively different from the Coulomb / S^3 case of Paper 0 [13] and Paper 7 [15], where the graph Laplacian $D - A$ on the (n, l, m) lattice converges to the continuum S^3 Laplace–Beltrami operator, whose eigenvalue $n^2 - 1$ on the n-th shell the universal kinetic constant $\kappa = -1/16$ converts to Rydberg energies. There the off-diagonal structure of A is essential to the spectrum. In the Bargmann graph, by contrast, the off-diagonal structure is spectroscopically informative but spectrally inert. This is the first key structural difference between the Coulomb and the harmonic oscillator discretizations, and it motivates the asymmetry analysis of Section IV.

IV. THE COULOMB / HO ASYMMETRY

The Bargmann graph and the Coulomb S^3 graph are both natural discretizations of central-potential quantum mechanics, but they differ in every structural feature except the use of angular momentum as a labeling system. Table I summarizes the comparison.[**?**]

A. The causal chain

The asymmetry is not a list of independent observations. It has a single causal structure in which each step is forced by the previous one.

1. *Geometry choice.* SO(4) symmetry of the Coulomb problem (the Runge–Lenz vector closes with the angular momenta into $\mathfrak{so}(4)$) makes S^3 the natural manifold, by Fock's reduction [1]. SU(3) symmetry of the harmonic oscillator (the ladder operators $a_i = (x_i + ip_i/m\omega)\sqrt{m\omega/2\hbar}$ transform as the fundamental **3** and generate $\mathfrak{su}(3)$ under commutation) makes $\mathbb{C}^3$ and its quotients $\mathbb{CP}^2$ and S^5 natural, by [6].

2. *Operator order.* A real Riemannian metric on S^3 makes the scalar Laplace–Beltrami operator (second-order in derivatives) the natural object. A complex Cauchy–Riemann or Kähler structure on S^5 or $\mathbb{CP}^2$ makes the Dolbeault, Kohn, or Euler operators (first-order in $\hat{N}$ on holomorphic sections) natural. This is not a free choice; it follows from the requirement that the operator be invariant under the dynamical symmetry group acting holomorphically on the manifold's structure.

3. *Spectrum growth.* A second-order self-adjoint operator on a compact Riemannian manifold has eigenvalues that grow polynomially with shell index, typically as n^2 for round spheres (Weyl asymptotics). A first-order equivariant operator on the symmetric irreducible representations of a compact Lie group has eigenvalues linear in the highest weight. These growths are different by definition.

4. *Projection type.* Mapping a quadratic mathematical eigenvalue $\lambda_n = n^2 - 1$ to an inverse-square physical energy $E_n = -Z^2/(2n^2)$ is a *nonlinear* operation: $E = -Z^2/[2(\lambda + 1)]$. Mapping a linear mathematical eigenvalue $\lambda_N = N$ to a linear physical energy $E_N = \hbar\omega(N + 3/2)$ is a *linear-affine* operation.

5. *Spectral zeta and boundary corrections.* A nonlinear projection on a compact manifold requires regularization at the boundary of the spectrum, captured by spectral zeta values $\zeta_M(s) = \sum_n \lambda_n^{-s}$. For round S^d these involve π through Bernoulli numbers and Riemann zeta values at integer arguments. A linear-affine projection has no such regularization: every term in the spectrum is finite, the sum is unconditionally convergent (or trivially conditionally convergent with the additive constant absorbed), and no transcendental seed appears.

6. *Calibration exchange constant.* The transcendentals in the spectral zeta are inherited by the physical spectrum as what Paper 18 [17] terms *calibration* exchange constants. They are the cost of going from the dimensionless graph to a physical energy via the nonlinear projection. For the harmonic oscillator this cost is zero; for the Coulomb problem it is π.

We summarize this causal chain as a structural observation:

Theorem 2 (Structural origin of calibration π)**.** *A quantum system whose spectrum is encoded by a self-adjoint graph Laplacian on a compact Riemannian manifold via a nonlinear projection requires a calibration exchange constant arising from the spectral zeta values of the manifold. A quantum system whose spectrum is encoded by a first-order equivariant operator on a compact complex manifold via a linear-affine projection requires no such calibration constant. The Coulomb potential and the three-dimensional isotropic harmonic oscillator instantiate the two cases respectively.*

This is a structural description rather than a standalone theorem in the strict mathematical sense. Each link in the causal chain is established by independent classical results (Weyl asymptotics; Borel–Weil; Folland–Stein; Paper 18); the contribution of the present paper is to assemble them into a single statement that the role of π in the Coulomb projection is structurally tied to second-order Riemannian geometry, and that there is no analogous role for π in the harmonic oscillator's complex-analytic projection.

B. Refinement of the exchange constant taxonomy

Paper 18 [17] classified exchange constants into four types: *intrinsic* (internal to the discrete graph), *calibration* (from projecting graph → physical spectrum), *embedding* (from embedding the graph into a continuous space for measurement), and *flow* (from time-dependent dynamics).

The Bargmann graph supplies a worked example that refines this classification. The Coulomb S^3 graph has both calibration and embedding π: calibration π in Paper 2's $K = \pi(B + F - \Delta)$ [14], embedding π in the

TABLE I. Structural comparison of the Coulomb and harmonic oscillator discretizations. Each row is a feature; each column is a system. The asymmetry has a causal structure: the geometry choice (row 2) determines the operator order (row 3), which determines the spectrum growth (row 4), which determines the projection type (row 6), which determines whether transcendental constants are needed (row 9).

Property	Coulomb (S^3, Riemannian)	HO (S^5 Hardy, complex)
Symmetry group	SO(4)	SU(3)
Manifold	S^3	S^5 (holomorphic sector)
Natural operator	Laplace–Beltrami (2nd order)	Euler / Dolbeault (1st order)
Eigenvalue λ_N	$n^2 - 1$ (quadratic in n)	N (linear in N)
Shell degeneracy	n^2	$(N+1)(N+2)/2$
Projection $E(\lambda)$	$E_n = -Z^2/[2(\lambda+1)]$ (nonlinear)	$E_N = \hbar\omega(\lambda + 3/2)$ (linear-affine)
Projection constant	$p_0 = Z/n$ (state-dependent)	$\hbar\omega$ (universal)
Graph Laplacian role	carries spectrum via conformal $D - A$ identification	encodes transitions only
Transcendentals in projection	π (calibration exchange constant)	none
π-free graph	yes	yes
π in full pipeline	enters at projection	never enters

electron-electron $1/r_{12}$ Slater integrals and the radial normalization of hydrogenic functions. The harmonic oscillator graph has only embedding π (the π^{-3} Gaussian normalization), and that embedding π is never used by the graph itself.

The refinement is therefore: *calibration* exchange constants are tied to nonlinear projections from second-order Riemannian operators. They are not a generic feature of quantum discretization, but a specific feature of the Coulomb / S^3 construction (and, we expect, of any other system whose dynamical symmetry group naturally acts on a Riemannian manifold rather than a complex one).

C. Fourth layer: modular-Hamiltonian Pythagorean orthogonality

The footnote at the head of Section IV records the original three layers of the Coulomb/HO asymmetry: the spectrum-computing role of L_0 (Coulomb-specific), the calibration π (Coulomb-specific), and Wilson gauge with natural matter coupling (Coulomb-specific). Sprints L2-E + L2-F.1 (May 2026) identified a fourth layer at the modular-Hamiltonian / wedge level:

- Layer 1: spectrum-computing role of L_0 (Coulomb-specific).
- Layer 2: calibration π (Coulomb-specific).
- Layer 3: Wilson lattice gauge with natural matter coupling (Coulomb-specific, via Papers 25, 30; S^5 is gauge-only per Sprint ST-SU3).
- Layer 4: modular-Hamiltonian Pythagorean orthogonality on the hemispheric wedge (Coulomb-specific; S^5 Hardy sector has no clean analog).
- Layer 5: spectral-action gravity termination (Paper 51). The CH Dirac spectral action on S^{2m+1} terminates at exactly $(m+1)$ power-law terms; only S^3 ($m = 1$) gives *pure Einstein gravity* (2 terms: cosmological + EH, no R^2 or higher-curvature corrections). S^5 ($m = 2$) terminates at 3 terms (includes R^2 Gauss–Bonnet). The mechanism is the Bernoulli identity for odd polynomials at half-integer shifts.
- Layer 6 *(new)*: master Mellin engine chirality-parity selection rule (Sprint Q5'-S5-falsifier, June 2026; `debug/sprint_q5p_s5_falsifier_memo.md`). On the Camporesi–Higuchi S^3 spectral triple at every finite $n_{\max}$, the Mellin source $\mathrm{Tr}(D^k\, e^{-tD^2})$ exhibits a bit-exact $\mathbb{Z}/2$ chirality-parity factorization (Sprint Q5'-CH-1): plain trace vanishes identically at odd k, supertrace vanishes identically at even k. On the Bargmann–Segal Hardy sector at every finite $N_{\max}$ this rule fails on both load-bearing structural conditions: the Hardy spectrum is strictly positive (no $\pm$-partner) and no half-integer-weight chirality grading exists on the $(N, 0)$ tower. This propagates the Coulomb/HO asymmetry from the modular-Hamiltonian level (Layer 4) and the spectral-action level (Layer 5) to the master Mellin engine level, naming a CH-substrate-specific structural prerequisite for the cosmic-Galois U^* enrichment program (Paper 55 § subsec:open_m2_m3).

The fourth-layer statement is structural rather than numerical. On the truncated Camporesi–Higuchi spectral triple at signature $(3, 1)$ (the Lorentzian Krein-space lift of the S^3 Coulomb discretisation; Paper 43 [25]), the Bisognano–Wichmann-aligned local Hamiltonian $H_{\mathrm{local}} := K_L^{\alpha,W}/\beta$ and the wedge-restricted Lorentzian Dirac D_W^L live in mutually-orthogonal Hilbert–Schmidt subspaces of the wedge operator algebra:

$$\langle H_{\mathrm{local}}, D_W^L \rangle_{\mathrm{HS}} \;=\; 0$$

bit-exactly at every tested $n_{\max}$ and N_t, across all six modular witnesses (Bisognano–Wichmann + Hartle–Hawking × 2 + Sewell + Unruh × 2). This holds because H_{local} is diagonal in the unfolded wedge spinor basis (eigenvalues two_m$_j > 0$, the rotation generator around the Hopf axis) and D_W^L is off-diagonal in the same basis (it intertwines $\pm m_j$ chirality partners via the wedge basis change). The closed-form residual lives entirely in the master Mellin engine M1 ring (Paper 32 § VIII):

$$\|H_{\mathrm{local}} - D_W^L\|_F^2 \;=\; \kappa_g^2 \cdot \frac{S(n_{\max})}{4\pi^2} \;+\; D(n_{\max}),$$

with S and D pure rationals in $n_{\max}$ (cumulative wedge Casimir traces).

a. Why the fourth layer fails on the S^5 Hardy sector. The Pythagorean structure requires the S^3 confluence of first-order D_{GV} *and* second-order Δ_{LB} coexisting on a parallelizable spinor bundle with half-integer Hopf-axis wedge. On the Bargmann–Segal Hardy sector $H^2(S^5)$ this confluence is structurally absent. Four named obstructions:

1. *No spinor sector on the $(N,0)$ tower.* The symmetric irreducible representations of SU(3) are bosonic (integer m_l); the Hardy sector has no half-integer weight content, hence no spinor lift compatible with the Bargmann transform.
2. *No second-order vs. first-order distinction with a separate Dirac.* The HO's spectrum-computing role is played by the first-order Euler operator $\hat{N} + 3/2$ on the Hardy sector, which *is* the Hamiltonian. There is no "D_W^L" sitting beside a Laplacian the way D_{GV} sits beside Δ_{LB} on S^3; the operands of the Pythagorean comparison collapse to two diagonal operators in the same subspace, and the inner product is generically nonzero.
3. *No half-integer wedge.* The natural BW-analog generator $K^{\alpha,W} = \mathrm{diag}(2m_l > 0)$ has *even*-integer spectrum, so the modular period closes at $\beta = \pi$ rather than $\beta = 2\pi$. The four-witness Wick-rotation theorem at $\beta = 2\pi$ is half-integer-spectrum specific.
4. *Coulomb/HO category mismatch resurfaces.* The layer-1/2/3 asymmetry already documented above propagates to the modular-Hamiltonian level: every structural ingredient Pythagorean orthogonality requires (spinor bundle + half-integer wedge + second-order Δ alongside first-order D) is the Coulomb-side outcome of layers 1–3.

These four obstructions are the same blocker that prevents the cross-manifold tensor product $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$ at the AC-extension level (Paper 32 § VIII.C G4b), now showing up at the modular-Hamiltonian level as a sibling obstruction rather than a new one. The six-layer Coulomb/HO asymmetry is consistent across spectrum computation, calibration, gauge-matter coupling, modular structure, spectral-action gravity termination, and master-Mellin chirality factorization.

b. Honest scope. The HS-orthogonality is empirically demonstrated and structurally explained via the diagonal / off-diagonal subspace decomposition. A formal proof of the subspace decomposition at general $(n_{\max}, N_t)$ is a named follow-on; the bit-exact numerical evidence at $n_{\max} \in \{1, \ldots, 5\}$ and across the 18-cell six-witness panel is strong evidence the orthogonality is structural, not coincidental. Supporting memos: `debug/h_local_residual_pslq_memo.md`, `debug/six_witness_hs_orthogonality_memo.md`, `debug/pythagorean_extension_scoping_memo.md`.

V. THE HARMONIC OSCILLATOR RIGIDITY THEOREM

The Fock projection rigidity theorem of Paper 23 [19] states that the S^3 Fock projection $p_0 = \sqrt{-2E_n}$ maps a one-electron central-field Hamiltonian onto the free Laplacian on S^3 if and only if the spectrum is l-degenerate within each n-shell, which is unique to the Coulomb potential by SO(4) symmetry.

The harmonic oscillator analog has the same structural form:

Theorem 3 (Bargmann–Segal rigidity for the 3D HO)**.** *The three-dimensional isotropic harmonic oscillator $V(r) = \frac{1}{2}m\omega^2 r^2$ is the unique central-field potential whose bound-state spectrum arises from the Euler operator on the Hardy space $H^2(S^5)$ restricted to the symmetric irreducible representations $(N,0)$ of* SU(3).

Proof. The argument has three independent classical inputs.

Step 1 (Jauch–Hill, 1940; uniqueness sharpening Wybourne / Iachello). Jauch and Hill [6] proved the forward direction: the three-dimensional isotropic harmonic oscillator possesses an SU(3) dynamical symmetry, constructing the explicit $\mathfrak{su}(3)$ generators as bilinears in the position and momentum operators (the angular momentum $\mathbf{L}$ and the quadrupole tensor $Q_{ij} = \frac{1}{2}(p_i p_j / m\omega + m\omega x_i x_j)$) and showing that they close into $\mathfrak{su}(3)$ under commutation. The converse — that $V(r) \propto r^2$ is the *only* central-field potential whose generators close into $\mathfrak{su}(3)$, so that the closure *forces* $V(r) \propto r^2$ — is the uniqueness sharpening established in the later dynamical-symmetry literature; we take the biconditional

$$V(r) = \tfrac{1}{2}m\omega^2 r^2 \iff \text{SU(3) dynamical symmetry}$$

in the form given by Wybourne [8], Chapter 16, and Iachello [9], Chapter 4.

Step 2 (Borel–Weil). The Borel–Weil theorem [7] identifies the symmetric irreducible representations $(N,0)$ of

SU(3) with the spaces of holomorphic sections of the line bundle $\mathcal{O}(N) \to \mathbb{CP}^2$. By restriction along the fibration $S^1 \to S^5 \to \mathbb{CP}^2$ (Eq. (8)), this is the same data as the homogeneous degree-N subspace $\mathcal{H}_N$ of the Hardy space $H^2(S^5)$ (Eq. (7)). Hence the $(N, 0)$ irreducible decomposition of an SU(3)-equivariant Hilbert space coincides exactly with the Hardy-space decomposition by holomorphic degree.

Step 3 (Bargmann–Segal–Hall). The Bargmann transform [2], made rigorous as a unitary intertwiner of group representations by Hall [4], establishes a unique (up to phase) unitary equivalence between $L^2(\mathbb{R}^3)$ with the Schrödinger representation of the harmonic oscillator and $F^2(\mathbb{C}^3)$ with the Bargmann representation, under which the harmonic oscillator Hamiltonian maps to $\hbar\omega(\hat{N}+3/2)$ (Eq. (5)). The Euler operator $\hat{N}$ is the unique self-adjoint operator on $F^2(\mathbb{C}^3)$ that is diagonal on the monomial basis with eigenvalue $a+b+c$ on $z_1^a z_2^b z_3^c$.

Combination. Suppose $V(r)$ is a central-field potential whose bound-state spectrum arises from the Euler operator on the Hardy-space sectors $\mathcal{H}_N$, with each $\mathcal{H}_N$ carrying the $(N, 0)$ irreducible representation of SU(3). By Step 2, the Hilbert space of $V(r)$ then decomposes under SU(3) into the symmetric irreducible representations $(N, 0)$, one per N, and the Hamiltonian is linear in $\hat{N}$ on each. By Step 3, this is exactly the image of the harmonic oscillator under the Bargmann transform. By the uniqueness of the Bargmann intertwiner (up to phase) and of the Schrödinger representation, the original Hamiltonian must be unitarily equivalent to the harmonic oscillator Hamiltonian. By Step 1, the only central-field potential with this property is $V(r) = \frac{1}{2}m\omega^2 r^2$, completing the converse. $\square$ $\square$

a. Remark on the converse step. The crucial input to the converse direction is the biconditional sharpening of Step 1 [8, 9]: the algebraic closure of the angular momentum and quadrupole generators into $\mathfrak{su}(3)$ *forces* $V \propto r^2$ (the forward implication is the original Jauch–Hill result [6]). This is the HO analog of the corresponding biconditional Fock–Bargmann result for the Coulomb problem (the Runge–Lenz closure forces $V \propto -1/r$). Both biconditional results are standard classical group theory [8, 9].

The harmonic oscillator and the Coulomb problem are therefore both rigid: the natural compact manifold and the natural operator together uniquely determine the physical potential through the closure conditions of the dynamical symmetry algebra. These two rigidity results are independent (Coulomb uses S^3 and Laplace–Beltrami; HO uses S^5 Hardy space and the Euler operator) and do not derive from each other, but they have parallel structures and parallel proofs. Together they exhaust the two potentials that classical group theory identifies as having full SO(4) and SU(3) dynamical symmetries respectively.

A. Two-fermion entanglement rigidity (corollary)

The HO rigidity theorem of Section V has a direct quantum-information corollary established empirically in Paper 27 [20]: *the spatial 1-RDM von Neumann entropy of the closed-shell two-fermion ground state on the Bargmann–Segal graph is identically zero for any central two-body interaction $V(r_{12})$.*

The mechanism is the Moshinsky–Talmi (MT) lab-to-relative transformation [21]: any central $V(r_{12})$ on the 3D HO basis preserves the total HO quantum number $N_{\text{tot}} = N_{\text{rel}} + N_{CM}$. The two-body Hamiltonian therefore commutes with the one-body Hamiltonian H_{HO} at the operator level (both are diagonal in N_{tot}), and the closed-shell ground state of $H_{HO} + V$ lies entirely within the lowest N_{tot} sector. For two fermions in the $M_L = 0, M_S = 0$ singlet at the lowest shell ($N_{\text{tot}} = 0$), that sector is one-dimensional — a single Slater determinant $|(0s)^2\rangle$ — so the spatial 1-RDM has occupation $(2, 0, \ldots, 0)$ exactly and the von Neumann entropy is zero. Numerical verification at $N_{\max} \in \{2, 3\}$ with the Minnesota NN singlet potential at $\hbar\omega = 10$ MeV gives $S_{\text{full}} = 0$ and $\|[H_{HO}, V]\|_F / \|H_{HO}\|_F < 10^{-15}$ (Paper 27, Sec. VII.A; [20]).

This is the structural dual of the corresponding statement for the Coulomb S^3 basis, where Paper 27 establishes a universal but *nonzero* two-variable scaling $S_B = A(\tilde{w}_B/\delta_B)^\gamma$ with $\gamma \to 2$ at large $Z, n_{\max}$. The Coulomb result is nontrivial because $V_{ee} = 1/r_{12}$ does not preserve any HO-like total-quanta quantum number on S^3; it couples across shells, generating the second-order Rayleigh–Schrödinger entanglement amplitude that Paper 27 measures. The HO basis kills this amplitude at the operator level by the MT conservation law, completing the universal/Coulomb-specific partition stated in the abstract: the HO graph is π-free in eigenvalues, in the Bargmann measure, and now also in two-fermion entanglement.

B. Spinor-Lichnerowicz corollary: the operator-order discriminant

The HO rigidity theorem of Section V and the Coulomb/HO asymmetry of Section IV established that calibration π (even-zeta content) enters from second-order Riemannian operators on scalar bundles: the HO via the Bargmann–Segal Euler operator on $H^2(S^5)$, and the Coulomb system via the Fock projection to $-\Delta_{\text{LB}}$ on S^3. The first-order/second-order distinction was identified as the structural origin of the π-free property of the Bargmann graph (first-order, complex-analytic, linear projection) versus the calibration π of the Fock Dirichlet series (second-order, Riemannian, nonlinear projection).

The Dirac-on-S^3 Tier 3 sprint (Track T9) extends this result to spinor bundles. The squared Dirac operator on unit S^3 satisfies the Lichnerowicz identity $D^2 = \nabla^*\nabla + R/4$ [22], where $R = 6$. Its spectral zeta evaluates at every integer s to a two-term polynomial in π^2 with rational coefficients (Paper 18, Eq. (T9)), with *no odd-zeta content* at any s.

The mechanism is algebraic: squaring maps the first-order Dirac sum $\sum g_n \cdot (\text{odd})^{-s}$ (which produces $\zeta_R(\text{odd})$ at odd s, as in Tier 1 Track D3) to the second-order sum $\sum g_n \cdot (\text{odd})^{-2s}$, which is always $\mathbb{Q} \cdot \pi^{2s}$ by the Bernoulli formula. The odd-zeta content of D is structurally lost upon squaring.

Corollary 1 (Operator-order transcendental discriminant)**.** *The operator-order distinction (first-order vs. second-order) is bundle-independent on round S^3. All second-order operators—scalar Laplace–Beltrami, spinor connection Laplacian $\nabla^*\nabla$, squared Dirac D^2—produce exclusively π^{even} spectral-zeta content. All first-order operators (Dirac D) can produce odd-zeta content ($\zeta_R(3), \zeta_R(5), \ldots$) via half-integer eigenvalue sums. The bundle type (scalar vs. spinor) modulates rational coefficients and the eigenvalue/charge lattice (integer vs. half-integer) but does not change the transcendental class.*

This corollary completes the universal/Coulomb-specific partition stated in the abstract: the Bargmann graph is π-free because it encodes a first-order operator (the Euler operator); calibration π is second-order; and this pattern persists on spinor bundles. Cross-reference: Paper 18 §IV (taxonomy closure); Tier 1 Track D3 ($\zeta(3)$ from first-order Dirac); T9 debug memo (`debug/dirac_t9_memo.md`).

C. Gauge-structure corollary: reduced Wilson content on S^5

Paper 25 [23] observes that the GeoVac S^3 Coulomb graph at finite $n_{\max}$ is simultaneously a Fock discretization of S^3, a triangle-free simplicial 1-complex with discrete Hodge decomposition, and an abelian Wilson-type $U(1)$ lattice gauge structure with matter on nodes and a $U(1)$ connection on edges. The same three structural layers are available on the Bargmann–Segal graph of the present paper, but with reduced physical content. This corollary documents the transfer pattern.

a. Positive transfer (abelian $U(1)$ Wilson–Hodge structure). The Bargmann graph at $N_{\max} = 5$ has 56 nodes, 165 edges, one connected component, and Euler characteristic $V - E = -109$. The signed incidence matrix $B \in \mathbb{Z}^{56\times 165}$, oriented along the creation-operator direction ($N \to N + 1$), gives a discrete exterior derivative $d_0 = B^\top$ with node Laplacian $L_0 = BB^\top$ and edge Laplacian $L_1 = B^\top B$. The first Betti number is $\beta_1 = E - V + c = 110$, counting the independent cycle classes that carry Wilson-loop holonomies in the abelian interpretation. Under a node-local $U(1)$ gauge transformation $\psi_v \mapsto e^{i\chi_v}\psi_v$, the dipole edge amplitudes $\langle w \,|\, z_q \,|\, v\rangle$ transform covariantly as $U_{vw} \mapsto e^{-i\chi_v} U_{vw} e^{i\chi_w}$. The combinatorial Hodge decomposition $\mathbb{R}^{165} = \operatorname{im} d_0 \oplus \ker L_1$ and the SVD identity between nonzero spectra of L_0 and L_1 both hold. At this level of abstraction, the Bargmann graph is a lattice gauge structure in exactly the Paper 25 sense.

b. Reduced content (no spectrum-computing role for L_0). On S^3, the node Laplacian $L_0 = D - A$ is conformally identified with the continuum S^3 Laplace–Beltrami operator, whose eigenvalues $n^2 - 1$ are mapped to the Coulomb spectrum by the kinetic constant $\kappa = -1/16$. The Wilson dictionary in Paper 25 identifies L_0 with the matter propagator's inverse (covariant Laplacian $-\nabla^2 + m^2$), and this identification has spectral content: the matter spectrum is literally the spectrum of L_0 up to κ-scaling. On S^5, by contrast, the Bargmann node Laplacian is spectrally inert for the physical HO spectrum: the HO energies $\hbar\omega(N + 3/2)$ live in the diagonal (Eq. (15)), not in $L_0 = D - A$. The graph adjacency A encodes dipole transition amplitudes only, and Theorem 3 below shows that the node Laplacian L_0 computed from A plays no role in reconstructing the spectrum.

c. Reduced content (no Wilson SU(3) structure). The complex Hopf fibration $S^1 \to S^5 \to \mathbb{CP}^2$ of Eq. (8) suggests a potential non-abelian SU(3) gauge structure, with link variables valued in the fundamental rep **3** and node matter in the symmetric irreps $(N, 0)$. This does not materialize. The dipole operator z_q is the fundamental **3** and intertwines $(N, 0) \otimes (1, 0) \to (N + 1, 0)$ as a Clebsch–Gordan coupling, not as a group element of SU(3). A Wilson lattice gauge structure requires the link variables to live in a fixed group (the same SU(3) on every edge) acting uniformly on a fixed matter irrep. The Bargmann graph has *different* SU(3) irreps on different shells (the representation dimension grows as $(N+1)(N+2)/2$), which is structurally a representation tower, not a lattice matter-gauge system. The natural gauge group of the Bargmann graph is therefore $U(1)$, inherited from the phase ambiguity of the node wavefunctions, not SU(3).

d. Reduced content (no $\mathbb{CP}^2$ base spectrum). The natural discrete Hopf quotient collapsing the m_l-fibers of the (SO(3)-adapted) Bargmann basis at fixed (N, l) gives a 12-sector graph at $N_{\max} = 5$:

$$(0,0),(1,1),(2,0),(2,2),(3,1),(3,3),$$
$$(4,0),(4,2),(4,4),(5,1),(5,3),(5,5)$$

with fiber dimensions $(2l + 1) = 1,3,1,5,3,7,1,5,9,3,7,11$. The scalar Laplace–Beltrami spectrum on $\mathbb{CP}^2$ with the Fubini–Study Kähler metric of constant holomorphic sectional curvature $K = 4$ is [26]

$$\lambda_k = 4k(k+2), \qquad k = 0, 1, 2, \ldots, \tag{16}$$

with degeneracy $(k + 1)^3$ (the dimension of the (k, k) SU(3) irrep). The 12-sector quotient Laplacian under the natural multiplicity weighting has eigenvalues $\{0, 2.22, 4.87, 4.94, 11.65, 12.11, 18.74, 28.09, 33.58, 50.61, 63.77, 99.43\}$. The best single-scale rescaling against the analytic CP^2 sequence $\{0, 12, 32, 60, 96, 140, \ldots\}$ has a maximum relative residual of 24.98%; the ratios $\lambda_{\mathrm{quot},k}/\lambda_{\mathrm{CP}^2,k}$ vary between 0.08 and 0.19 across the 12 pairs, with no uniform rescaling. The m_l-quotient is therefore a combinatorial SO(3)-labeled graph on 12 sectors, not a spectral discretization of $\mathbb{CP}^2$. At this $N_{\max}$, there are simply too few sectors to represent the four-dimensional $\mathbb{CP}^2$ Laplacian, and the weighting scheme in the quotient is inherited from SO(3) branching, not from the Kähler volume form on $\mathbb{CP}^2$.

Corollary 2 (Gauge-structure reduction on the Bargmann graph)**.** *The abelian $U(1)$ Wilson–Hodge structure of the S^3 Coulomb graph (Paper 25) transfers to the Bargmann–Segal S^5 graph at the combinatorial level (incidence matrix, node/edge Laplacians, discrete Hodge decomposition, β_1 Wilson-loop classes), but the physical content of the structure is strictly reduced:*

1. *The node Laplacian $L_0 = D - A$ carries the Coulomb spectrum on S^3 through its conformal identification with the continuum Laplace–Beltrami operator ($\kappa = -1/16$) but is spectrally inert on S^5 (the HO spectrum is in the diagonal).*
2. *The natural gauge group of the graph is $U(1)$; no Wilson* SU(3) *lattice structure emerges from the $(N, 0)$ symmetric irreps, because transitions between distinct irreps are Clebsch–Gordan intertwiners, not group elements.*
3. *The m_l-fiber-collapse quotient does not produce a spectral discretization of $\mathbb{CP}^2$ at finite $N_{\max}$.*

The corollary extends the Coulomb/HO asymmetry of Theorem 2 from a two-layer statement (spectrum: yes S^3, no S^5; calibration π: yes S^3, no S^5) to a three-layer statement adding gauge-structure content (Wilson interpretability: full on S^3, reduced on S^5); this is the third of the six layers catalogued in §IV C. The universal content — that the Wilson–Hodge vocabulary applies combinatorially to any simplicial 1-complex — is preserved, but the Coulomb-specific content — that the graph Laplacian simultaneously carries the physical spectrum (through its conformal identification with the continuum Laplacian) and supports a nontrivial matter-gauge double — is not.

A practical consequence for the Paper 2 fine-structure-constant Observation [14]: the S^5 analog of the Paper 2 combination rule $K = \pi(B + F - \Delta)$ does not exist in natural form. A candidate F-analog (the Dirichlet series $\sum_{N\geq 1} g_N^{\mathrm{HO}}/N^{d_{\max}}$ with $g_N^{\mathrm{HO}} = (N + 1)(N + 2)/2$ and packing exponent $d_{\max} = \dim \mathbb{C}^3 = 6$) evaluates to $\frac{1}{2}[\zeta_R(4) + 3\zeta_R(5) + 2\zeta_R(6)]$, a mixed even/odd zeta combination with no clean $\zeta_R(2) = \pi^2/6$ projection; and Paper 23's Fock projection rigidity theorem establishes that the spectrum-computing projection with calibration π is unique to S^3 (not S^5), so the B and Δ analogs (both tied to the Coulomb projection) also do not lift. The S^5 graph is thus a lattice gauge structure with abelian $U(1)$ content but without α-like spectral content, consistent with the operator-order transcendental discriminant of Section V B. Cross-reference: Paper 25 §VII.1 (the open question answered here); debug memo `debug/s5_gauge_structure_memo.md`; numerical data `debug/data/s5_graph_spectrum.json`.

VI. IMPLICATIONS

A. For the fine-structure constant Observation

Paper 2 [14] observes that the fine-structure constant satisfies the cubic equation $\alpha^3 - K\alpha + 1 = 0$ with $K = \pi(B+F-\Delta)$, where $B = 42$, $F = \zeta(2) = \pi^2/6$, $\Delta = 1/40$ are spectral invariants of the Hopf fibration $S^1 \to S^3 \to S^2$. The numerical value $K = 137.036064$ matches $1/\alpha$ to 8.8×10^{-8}, with a combinatorial p-value of 5.2×10^{-9}. The combination rule $K = \pi(B + F - \Delta)$ is an empirical Observation.

The Coulomb / HO asymmetry developed in Section IV provides structural support for one element of the formula: the prefactor π. By Theorem 2, calibration π of this type is tied to second-order Riemannian operators with nonlinear projections. The harmonic oscillator's complex / first-order construction has no such π. Therefore π in $K = \pi(B + F - \Delta)$ is structurally Coulomb-specific: it comes from the spectral zeta values of S^3, which exist because the Coulomb projection $E_n = -Z^2/(2n^2)$ is nonlinear, and there is no analog in any other dynamical-symmetry-classified central potential.

This does not derive Paper 2's combination rule. It explains why π appears in K, but does not explain the additive combination $B + F - \Delta$ or the values of B, F, Δ themselves. Paper 2's combination rule remains an Observation; its status is unchanged by this paper. The contribution here is to sharpen the framing: any future derivation of the combination rule must rely on the SO(4) / Riemannian features of S^3 that distinguish the Coulomb problem from every other central potential, not on the broader SO(3) angular momentum structure that S^3 shares with S^5 and other manifolds.

B. For nuclear quantum simulation

Paper 23 [19] constructs nuclear shell-model qubit Hamiltonians on the harmonic oscillator single-particle basis, producing the deuteron and helium-4 qubit Hamiltonians via the Minnesota nucleon-nucleon potential and Moshinsky–Talmi brackets. Theorem 1 of the present paper implies that the single-particle structure of these Hamiltonians is exactly rational at the graph level: the diagonal HO energies, the orbital labeling, and the dipole transition selection rules are all built from rational arithmetic on integers. Any floating-point error in nuclear Pauli coefficients comes from the nucleon–nucleon interaction matrix elements (which involve Gaussian integrals via Moshinsky–Talmi brackets), not from the single-particle basis or the graph topology. This is a precision guarantee for nuclear qubit Hamiltonians built within the GeoVac framework.

C. The universal / Coulomb-specific partition completed

Combined with Paper 22 [18] (potential-independent angular sparsity) and Paper 23 [19] (Fock projection rigidity theorem), the present paper completes a three-way partition of the GeoVac framework's claims:

a. Universal across spherical fermion systems. The angular-momentum selection rules from Wigner $3j$ symbols and the Gaunt integral; the ERI density bound (1.44% at $l_{\max} = 3$, independent of $V(r)$); the block-diagonal structure of two-body operators; and the $O(Q^{2.5})$ composed Pauli scaling. These are the Paper 22 results.

b. Coulomb-specific. The S^3 Fock conformal projection; the calibration exchange constant π; the Hopf bundle structure underlying the α Observation; the kinetic constant $\kappa = -1/16$ from the S^3 Laplacian. These are the Paper 23 / Track NH results and the present paper.

c. Harmonic-oscillator-specific (this paper). The S^5 Hardy-space discretization; the linear-affine projection $E_N = \hbar\omega(N + 3/2)$; the bit-exactly π-free graph at every $N_{\max}$; the embedding π^{-3} in the continuous Bargmann measure that the discrete graph never invokes; the Bargmann–Segal rigidity theorem (Theorem 3).

The two natural geometries (S^3 for Coulomb and S^5 Hardy for HO) are not in competition; they are the discrete encodings of two physically distinct potentials with two distinct dynamical symmetry groups. The asymmetry between them is structural and informs which features of the GeoVac framework extend to nuclear, atomic, and molecular systems beyond the electronic Coulomb problem.

VII. CONCLUSION

We constructed a discrete graph encoding of the three-dimensional isotropic harmonic oscillator on the holomorphic sector of S^5, proved that the graph and its projection to physical energies are bit-exactly π-free in exact rational arithmetic (Theorem 1), and developed the structural origin of this fact in terms of the difference between Riemannian and complex natural geometries (Theorem 2). We stated and proved the harmonic oscillator analog of the Fock projection rigidity theorem (Theorem 3), based on the biconditional uniqueness of SU(3) dynamical symmetry (forward direction due to Jauch–Hill, converse to Wybourne and Iachello), the Borel–Weil identification of $(N, 0)$ irreducibles with the holomorphic Hardy-space sectors, and the Bargmann–Segal–Hall unitary intertwiner.

Calibration π is Coulomb-specific. It is the cost of mapping a quadratic Riemannian eigenvalue to an inverse-square physical energy, and it has no analog for the harmonic oscillator's linear complex-analytic projection. This refines Paper 18's exchange constant taxonomy and sharpens Paper 2's framing of the fine-structure constant Observation: any geometric derivation of α must rely on the SO(4) / Riemannian features of the Coulomb problem, not on the angular momentum structure shared with other central potentials.

Future work includes extending the discretization to anisotropic harmonic oscillators (which break SU(3) to a smaller subgroup, with corresponding modifications of the manifold), and identifying the natural compact manifolds for other dynamical-symmetry-classified central potentials such as the Morse oscillator (SU(1, 1)), the Kepler problem in n dimensions (SO($n + 1$)), and the Pöschl–Teller hierarchy. Each such system should be classifiable by whether its natural operator is first-order (no calibration exchange constant) or second-order (calibration exchange constant required), providing a structural taxonomy of where transcendental constants enter the discrete-to-physical map.

[1] V. Fock, "Zur Theorie des Wasserstoffatoms," *Z. Phys.* **98**, 145 (1935).

[2] V. Bargmann, "On a Hilbert space of analytic functions and an associated integral transform," *Comm. Pure Appl. Math.* **14**, 187 (1961).

[3] I. E. Segal, "Mathematical problems of relativistic physics," in *Proc. Summer Sem. Boulder, Colorado, 1960* (American Mathematical Society, Providence, 1963).

[4] B. C. Hall, "The Segal–Bargmann 'coherent state' transform for compact Lie groups," *J. Funct. Anal.* **122**, 103 (1994).

[5] G. B. Folland and E. M. Stein, "Estimates for the $\bar{\partial}_b$ complex and analysis on the Heisenberg group," *Comm. Pure Appl. Math.* **27**, 429 (1974).

[6] J. M. Jauch and E. L. Hill, "On the problem of degeneracy in quantum mechanics," *Phys. Rev.* **57**, 641 (1940).

[7] A. Borel and A. Weil, "Représentations linéaires et espaces homogènes kählériens des groupes de Lie compacts," as exposited in J.-P. Serre, *Représentations linéaires et espaces homogènes Kähleriens des groupes de Lie compacts* (Séminaire Bourbaki, exposé 100, 1954). Modern textbook treatment in W. Fulton and J. Harris, *Representation Theory: A First Course* (Springer, 1991), Lecture 23.

[8] B. G. Wybourne, *Classical Groups for Physicists* (Wiley-Interscience, New York, 1974), Chapter 16.

[9] F. Iachello, *Lie Algebras and Applications* (Springer, Berlin, 2006), Chapter 4.

[10] D. J. Rowe, "Microscopic theory of the nuclear collective model," *Rep. Prog. Phys.* **48**, 1419 (1985).

[11] A. R. Edmonds, *Angular Momentum in Quantum Mechanics* (Princeton University Press, 1957).

[12] D. A. Varshalovich, A. N. Moskalev, and V. K. Khersonskii, *Quantum Theory of Angular Momentum* (World Scientific, Singapore, 1988).

[13] J. Loutey, "Angular Momentum as Information Geometry: Isotropic Packing and the Universal Angular Cross-Section," GeoVac Paper 0 (2026).

[14] J. Loutey, "The Fine Structure Constant as a Spectral Coincidence on the Hopf Fibration," GeoVac Paper 2 (2026). [Observation.]

[15] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology," GeoVac Paper 7 (2026).

[16] J. Loutey, "Structurally Sparse Qubit Hamiltonians from Spectral Graph Theory," GeoVac Paper 14 (2026).

[17] J. Loutey, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," GeoVac Paper 18 (2026).

[18] J. Loutey, "Potential-Independent Angular Sparsity for Qubit Hamiltonians in Spherical Fermion Systems," GeoVac Paper 22 (2026).

[19] J. Loutey, "Nuclear Shell Model Hamiltonians on the Hyperspherical Lattice," GeoVac Paper 23 (2026).

[20] J. Loutey, "Entropy as a Projection Artifact: One-Body Operators are Entanglement-Inert on Sparse Lattices," GeoVac Paper 27 (2026).

[21] M. Moshinsky and Yu. F. Smirnov, *The Harmonic Oscillator in Modern Physics* (Harwood Academic, 1996).

[22] A. Lichnerowicz, *Spineurs harmoniques*, C. R. Acad. Sci. Paris **257**, 7–9 (1963).

[23] J. Loutey, *The Hopf Graph as a Lattice Gauge Structure: A Framework Observation*, Paper 25 of the Geometric Vacuum series (synthesis, 2026).

[24] J. Loutey, *The GeoVac spectral triple*, Paper 32 of the Geometric Vacuum series (synthesis, 2026).

[25] J. Loutey, *Lorentzian extension of the four-witness Wick-rotation theorem at finite cutoff*, Paper 43 of the Geometric Vacuum series (standalone, 2026).

[26] M. Berger, P. Gauduchon, and E. Mazet, *Le Spectre d'une Variété Riemannienne*, Lecture Notes in Mathematics **194**, Springer (1971); see also Ch. 5 for the scalar Laplace spectrum on $\mathbb{CP}^n$ with the Fubini–Study metric.

[27] V. Higgs and D. Pickrell, "Spherical Harmonic Oscillators," arXiv:2503.23549 (2025).